

2021 Exam Paper

Question 1

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- (i) Semi-group
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Question 4

- (a)(i) Draw this graph
- (a)(ii) Does (V, E) contain a complete subgraph?
- (a)(iii) What is the minimal number of edges you would have to remove to make this graph disconnected? Justify your answer.
- (a)(iv) What is the minimal number of edges you would have to remove to make this graph regular? Justify your answer
- (a)(v) Does this graph have an Eulerian trail? Justify your answer
- (a)(vi) Does this graph have a Hamiltonian circuit? Justify your answer.
- (a)(viii) Is this graph a tree?

(b)Isomorphism

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Question 5

(a) Is the set of all natural numbers divisible by 4 finite, countably infinite, or uncountably infinite? Justify your answer.

(b) Let A = {(x, y) ∈ R^2 | x + 2y = 1}. Is A finite, countably infinite, or uncountably infinite? Justify your answer.

(c) Let A = {(x, y) ∈ R^2 | x^2 + y^2 < 25} ∩ Z^2. Is A finite, countably infinite, or uncountably infinite? Justify your answer.

(d) In lecture we defined the language EQ_TM = {⟨M1, M2⟩ M1 and M2 are Turing machines and L(M1) = L(M2)}. Is EQ_TM finite, countably infinite, or uncountably infinite? Justify your answer.

Question 6

(a) Consider the language over the binary alphabet A = {0, 1} given by L={01^m01^m | m∈N, m≥0}. Use the Pumping Lemma to show the language L is not regular.

(c) Let L and M be two Turing-decidable languages over a finite alphabet A.Write down an enumerator that outputs the language

Question 1

Part a

$$let\ F\ [0,1] \to \mathbb{R} \text{ define by } f(x) = 2x^3 + 3 \ \forall \in [0,1]$$

one to one relationship (Injective)

$$let\ x_1, x_2 \in [0,1], f(x_1) = f(x_2), x_1 = x_2$$
$$Assume\ f(x_1) = f(x_2)$$
$$2x_1^3 + 3 = 2x_2^3 + 3$$
$$x_1^3 = x_2^3$$
$$x_1^3 - x_2^3 = 0$$
$$(x_1 - x_2)(x_1^2 + x_1x_2 + x_2^2)$$
$$: \text{only solution is } x_1 = x_2$$

if the deviation is always increasing (never change sign)

Alternative:

$$f'(x) = 6x^2$$
$$f'(0) = 0$$
$$f'(1) = 6$$
$$6x^2 \geq 0$$
$$=> f'(x) \text{ does not change sign on } (0, 1)$$

(onto) Surjective:

$$\forall y \in \mathbb{R}, \exists x \in (0, 1)\ s.t.\ f(x) = y$$

f is only increasing on (0, 1) ⇒ f(1) = max value of f = 5 (the maximum value), the R numbers contain numbers greater than 5.

⇒ f is not surjective.

Part b

$$Let\ B \text{ be the set of all nonempty regular languages over a finite alphabet A.}$$
$$\text{For L, M} \in B, L \sim M,\ iff\ L \cap M \neq \emptyset$$

Regular language = a language that accepted by a finite state acceptor.

(i) Reflexive if

$$L \sim L, \forall L \in B$$
$$Yes,\ because\ L \cap L \text{ which is nonempty by definition}$$

(ii) Symmetric if

$$if\ L \sim M => M \sim L,$$
$$yes,\ \text{because } L \cap M \neq \emptyset => M \cap L \neq \emptyset$$

(iii) Antisymmetric if:

$$if\ L \sim M\ and\ M \sim L, then\ M = L$$
$$no, e.g.\ L \cap L \cup \{w\} = \emptyset$$
$$where\ w \notin L$$
$$but\ L \neq L \cup \{w\}$$

(iv) Transitive if:

$$if\ L \sim M, M \sim N, then\ L \sim N$$
$$no, as, L, M, N \text{ may have distinct words in common}$$

(v) Equivalence relation if :

$$if \sim \text{ is reflexive, transitive and symmetric}$$
$$\sim \text{ is not transitive}$$
$$=> \sim \text{ is not an equivalence relation}$$

(vi) Partial order of:

$$if \sim \text{ is reflexive, transitive and antisymmetric}$$
$$\sim \text{is not transitive and antisymmetric}$$
$$=> \sim \text{is not a partial order}$$

Question 2

(i) Semi-group

If (C, o) is a semigroup then o is a binary associative operation on C

Is L o M in C, for all L, M in C?

L o M = { u o v | u in L, v in M }

u and how even length ⇒ u o v also has even length

L o M has only words with even lengths

The concatenation of two regular languages is still regular

since concatenation is closed on regular language L o M is regular ⇒ L o M in C

We also know that concatenation is associative operation on regular languages, ⇒ concatenations is an associative binary operation

⇒ (C, o) is a semigroup.

(ii) Monoid

A monoid is a semigroup with on identity element.

we need i in C, s.t. i o L = L = L o i

if we let i = {empty string}

i is finite so it is regular

the length of empty string = 0, so its even

⇒ i in C ⇒ (C, o) is a monoid.

(iii) Group

A group is a monoid where every element has an inverse.

For all L in C, we need M in C such that L o M = {empty string}

For all L with at least one non-empty w in L, there not exist M, such that L o M = {empty string}

C is not a group.

(iv) finite, countably infinite, or uncountably infinite?

set of all regular languages is countably infinite,

The set B of all regular language is countably infinite

But we know C is a subset of B

Let Li be in A, Li = a^2i, for all i in N, Li in C, there is an infinite number of Li's in C, C⇒ is not finite

C ⇒ countably infinite.

Question 3

N/A

Question 4

Let (V, E) be the undirected graph with vertices a, b, c, d, e, f, g, h, i, and j, and edges ab, ac, bc, bd, de, ce, df, fg, eg, dh, ej, bh, fh, f i, gi, gj, and cj.

(a)(i) Draw this graph

(a)(ii) Does (V, E) contain a complete subgraph?

b-d-h

(a)(iii) What is the minimal number of edges you would have to remove to make this graph disconnected? Justify your answer.

!(at least 2 distinct path) = The graph is connected so need at least one edge to be removed. But for every two vertices is V, there are least two distinct paths. Between them ⇒ one is insufficient .

2 remove a-b and a-c and the graph will become disconnected.

(a)(iv) What is the minimal number of edges you would have to remove to make this graph regular? Justify your answer

A graph (V, E) is called regular is $\exists k \in N$ s.t. (V, E) is k-regular. (All the edges have the same degree)

~~I don't think its possible, because:~~

Corollary: Let (V,E) be a k-regular graph. Then k#(V) = 2#(E) where #(V) is the number of vertices and #(E) is the number of edges.

~~and since the 2 * number of edge = 2 * 10 = 20 and we only have 17 edges, unless we remove all the edges for a degree of 0?~~

As the minimum degree of the graph is two, we need to remove at least enough edges of every edge two. In a two regular graph with 10 vertices there will be k#(V) / 2 = #(E) = 10

if there are only 10 edges we have to remove 7.

(a)(v) Does this graph have an Eulerian trail? Justify your answer

An Eulerian trail in a graph is a trail that traverses every edge of that graph. In other words, an Eulerian trail is a walk that traverses every edge of the graph exactly once. Trail ⇒ an edge is traversed at most once.

Eulerian ⇒ every edge is traversed.

You can repeat vertices

the degree of the all the vertices have to be even (Eulerian Circuit)

there is exactly 0 or 2, at most two vertices of odd degree.

Clear (V, E) satisfies this as h and j have degree 3 and the other have degree 2 and 4. Yes, Eulerian trail.

(a)(vi) Does this graph have a Hamiltonian circuit? Justify your answer.

A Hamiltonian circuit in a graph is a simple circuit that passes through every vertex of the graph.

Circuit that visits Every vertex exactly once.

Theorem: Every complete graph Kn for n ≥ 3 has a Hamiltonian circuit.
Proof: Let V = {v1, v2, v3, …vn} be the set of vertices of Kn, then v1v2v3…vnv1 is a Hamiltonian circuit. All edges in this circuit are part of Kn because Kn is complete.
qed

You can repeat vertices

You cannot repeat edges

There exist a 2 regular graph.

a-b-h-d-f-c-g-e-i-c-a

(a)(viii) Is this graph a tree?

A forest is an acyclic graph.

A tree is a connected acyclic graph / connected forest

There is a cycle in the graph a b c a

The graph is not a tree.

(b)Isomorphism

$$\phi : V \to V$$
$$\text{As a that presences the edges between vertices}$$
$$\forall x \in V, \phi(x) = x$$
$$\phi_2 : V \to V$$
$$\phi_2(a) = i$$
$$\phi_2(b) = f$$
$$\phi_2(c) = g$$
$$\phi_2(d) = d$$
$$\phi_2(e) = e$$
$$\phi_2(f) = b$$
$$\phi_2(g) = c$$
$$\phi_2(h) = h$$
$$\phi_2(i) = a$$
$$\phi_2(j) = j$$

(c) Prim

Question 5

(a) Is the set of all natural numbers divisible by 4 finite, countably infinite, or uncountably infinite? Justify your answer.

C.I, using bijection.

Surjective: Let y in A $\Rightarrow$ y = 4n = y = f(n)

(b) Let A = {(x, y) $\in$ R^2 | x + 2y = 1}. Is A finite, countably infinite, or uncountably infinite? Justify your answer.

U.I, only depends on R^2. Using bijection.

$$f : \mathbb{R}^2 \to A$$
$$f : \mathbb{R} = f(x, \frac{1}{2}(a - x))$$
$$SUR : let\ z \in A => y = (x, \frac{1}{2}(1 - x)) = f(x), x \in R$$

(c) Let A = {(x, y) ∈ R^2 | x^2 + y^2 < 25} ∩ Z^2. Is A finite, countably infinite, or uncountably infinite? Justify your answer.

$$A = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 < 25\} \cap \mathbb{Z}^2$$
$$= \{(x, y) \in \mathbb{Z}^2 | x^2 + y^2 < 25\} \text{ (set of integer within the circle of radius 5)}$$

Finite.

(d) In lecture we defined the language EQ_TM = {⟨M1, M2⟩ M1 and M2 are Turing machines and L(M1) = L(M2)}. Is EQ_TM finite, countably infinite, or uncountably infinite? Justify your answer.

$$EQ_{TM} = \{< M_1, M_2 > | L(M_1) = L(M_2)\}$$
$$A_{TM} = \{< M_1, M_2 >\} => EQ_{TM} \subseteq A$$
$$A \text{ is countably infinite}$$
$$B_{TM} = \{< M, M > | \text{ M is a TM}\}$$
$$\text{Clearly } B_{TM} \text{ is of one to one correspondence of all Turing Machine}$$
$$\text{so it is countably infinite}$$
$$EQ_{TM} \text{ is countably infinte}$$

Question 6

(a) Consider the language over the binary alphabet A = {0, 1} given by L={01^m01^m | m∈N, m≥0}. Use the Pumping Lemma to show the language L is not regular.

$$A = \{0, 1\} \; given \; by \; L = \{01^m01^m | m \in N, m \geq 0\}$$
$$\text{Assume L is regular, from the pumping lemma}$$
$$\text{there exist a pumping length p, } p \in \mathbb{N}$$
$$\text{such that } \forall w \in L \text{ with } |w| \geq p$$
$$w = 01^p01^p, \text{clearly } w \in L, |w| > p$$
$$=> w = xuy$$
$$=> xu = 01^a, a < p$$
$$=> \text{There are 2 cases for u:}$$

$$1.\ x = \epsilon => u = 01^a$$
$$2.\ x = 01^b => u = 1^c$$
$$Case\ 1 : w = 01^a1^{p-a}01^p$$
$$=> xu^2y = 01^a01^a1^{p-a}01^p \notin L =><=$$
$$Case\ 2 : w = 01^b1^c1^{p-b-c}01^p$$
$$=> xu^2y = 01^b1^{2c}1^{p-b-c}01^p = 01^{p+c}01^p =><=$$
$$\text{L is not regular}$$

(c) Let L and M be two Turing-decidable languages over a finite alphabet A.Write down an enumerator that outputs the language

$$L - M = \{u | u \in L\ and\ u \notin M\}$$

Explain why the enumerator you wrote down does indeed output all strings in L − M and no others

Turing-decidable = no loop, accept or reject. Enumerator ⇒ Print out an word of a Language

$$let\ M_1\ decide\ L$$
$$let\ M_2\ decide\ M$$
$$E = \text{ignore input}$$
$$A^* \text{ is countably infinite } => A^* = \{w_1, w_2....\}$$
$$\text{For i = 1,2... do the following:}$$
$$(i)\ \text{Run M}_1\ on\ w_i$$
$$\text{if ACCEPT go to (ii)}$$
$$(ii)\ Run\ M_2\ on\ w_i$$
$$\text{if REJECT, print } w_i$$